

D857

Secondary Earth Science Curriculum

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A first-year teacher at a suburban high school with a new planetarium is planning a three-week unit on weathering and erosion. She has gathered a dozen favorite hands-on activities and wants to start putting them in order. Following backward design, what should she do first?

- A. Define what students should know and be able to do, then decide what evidence will show it.
- B. Sequence the activities from easiest to hardest.
- C. Follow the textbook chapter order.
- D. Choose the activities that make the most use of the planetarium.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A unit plan lists five engaging labs, then says the final assessment “will be decided once we see how the labs go.” What is the main design flaw?

- A. The final assessment should be multiple choice.
- B. Five labs is too many for one unit.
- C. Goals and assessment evidence were not set first, so the labs may not line up with what students must learn.
- D. Labs should always come after lectures.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

An Earth science teacher, whose classroom has a large wall map, plans a lesson in which students analyze earthquake location data to find patterns that reveal plate boundaries and then explain those patterns using cause and effect. Her district adopted new laptops this year. Which description best fits this lesson?

- A. A content-only lesson, since the topic is plate tectonics.
- B. An inquiry lesson only if students design their own experiment.
- C. A three-dimensional lesson: a practice (analyzing data), a core idea (plate tectonics), and crosscutting concepts (patterns, cause and effect).
- D. A technology lesson, which is the same as three-dimensional learning.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

During a moon-phase lesson in a classroom with blackout curtains, a student explains that a crescent Moon happens because Earth's shadow covers most of the Moon. The class has been using a lamp and foam balls. What is the most accurate correction?

- A. Half the Moon is always sunlit; the phase shows how much of that lit half faces Earth. Earth's shadow matters only in a lunar eclipse.
- B. The student is right; phases are small lunar eclipses.
- C. Earth's shadow causes every phase except full moon.
- D. Phases happen because the Moon's distance from Earth changes.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

The Moon is between Earth and the Sun. What phase is seen from Earth?

- A. A solar eclipse, every time.
- B. New moon.
- C. First quarter.
- D. Full moon.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

For a unit on plate tectonics, a teacher with a popular rock collection writes a big-idea statement: "Plates move because ...". Which completion is accurate for a secondary course?

- A. Earthquakes push them apart.
- B. They float on a liquid mantle like rafts.
- C. Earth's rotation spins the plates across the surface.
- D. Heat escaping Earth's interior drives slow mantle convection, and gravity pulls cold, dense slabs down at subduction zones.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

While designing a water-cycle lesson for a school near a large reservoir, a teacher wants students to name the main energy source that drives evaporation and keeps the cycle running. The reservoir supplies 200,000 people. Which answer should the lesson build toward?

- A. Pumps at dams and reservoirs.
- B. Heat from Earth's interior.
- C. The Moon's gravity, through the tides.
- D. Energy from the Sun, with gravity returning water downhill.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A coastal district's ocean-science elective, which takes one field trip per semester, is scheduling a tide-pool visit to see the lowest low tides. The trip coordinator has a boat license. Which moon phase should the teacher choose?

- A. Any phase, since tides depend only on the Moon.
- B. Full moon only, because moonlight pulls on the water.
- C. First or third quarter moon.
- D. New or full moon, when Sun, Earth, and Moon line up and tidal range is greatest.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

During a first-quarter moon, what kind of tides occur?

- A. The highest tides of the month, because the Moon is closest.
- B. Neap tides, with the smallest range, because the Sun's and Moon's pulls partly offset.
- C. No tides, because only half the Moon is lit.
- D. Spring tides with the largest range.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A teacher, using a star-chart app the whole department likes, has students plot stars on an H-R diagram. A student notices that most plotted stars fall along a diagonal band. What should the teacher help students conclude?

- A. The band shows one star moving along it as it ages.
- B. Most stars fall there because stars spend most of their lives fusing hydrogen in their cores on the main sequence.
- C. Along the band, brighter stars are cooler.
- D. All stars in the band are the same age.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Compared with a main-sequence star at the lower right of the H-R diagram, a main-sequence star at the upper left is:

- A. Cooler, dimmer, and longer-lived.
- B. The same star later in its life.
- C. Hotter, more luminous, more massive, and shorter-lived.
- D. Hotter but less massive and longer-lived.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A team of teachers at a school with block scheduling is sequencing a ninth-grade Earth science course. Students have not yet studied density or heat transfer. The team wants to open the year with plate tectonics because a volcano documentary just came out. What is the best sequencing decision?

- A. Teach density and heat transfer first so students have the ideas needed to explain plate motion.
- B. Start with plate tectonics and skip density, since it belongs in physics.
- C. Follow the textbook order no matter what students already know.
- D. Always begin a course with space science because it is most engaging.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A ~~first-year teacher at a suburban high school with a new planetarium~~ is planning a three-week unit on weathering and erosion. She has gathered a dozen favorite hands-on activities and wants to start putting them in order. Following backward design, what should she do first?

Target word: first step in backward design **Decides:** Goals come before activities.

A	Correct	Backward design begins with desired results and assessment evidence.
B	Inverted operation	Starts with activities — the reverse order of backward design.
C	Adjacent category	Coverage order is not a design method tied to goals.
D	Camouflage	The facility does not set learning goals.

2. Answer C Concept 1 • transfer

A unit plan lists five engaging labs, then says the final assessment “will be decided once we see how the labs go.” What is the main design flaw?

Target word: flaw in the plan **Decides:** Assessment is chosen last.

A	Adjacent category	Format is a later choice; the problem is timing.
B	Wrong stem number	The number of labs is not the design flaw.
C	Correct	Backward design sets outcomes and evidence before activities.
D	Absolute language	Sequence of lab and lecture is not the issue.

3. Answer C Concept 2

An Earth science teacher, ~~whose classroom has a large wall map~~, plans a lesson in which students analyze earthquake location data to find patterns that reveal plate boundaries and then explain those patterns using cause and effect. ~~Her district adopted new laptops this year.~~ Which description best fits this lesson?

Target word: type of lesson design **Decides:** Practice + core idea + crosscutting concept.

A	Dropped element	Drops the practice and crosscutting concepts students use.
B	Absolute language	Analyzing existing data is a science practice.
C	Correct	All three dimensions are used together.
D	Camouflage	Laptops are a tool, not a dimension of learning.

4. Answer A Concept 3

During a moon-phase lesson ~~in a classroom with blackout curtains~~, a student explains that a crescent Moon happens because Earth’s shadow covers most of the Moon. ~~The class has been using a lamp and foam balls.~~ What is the most accurate correction?

Target word: correct explanation of phases **Decides:** Half the Moon is always lit.

A	Correct	Viewing geometry, not shadow, causes phases.
B	Confusion set	Confuses phases with eclipses.
C	Absolute language	Earth’s shadow causes no ordinary phase.
D	Adjacent category	Distance changes are small and do not create phases.

5. Answer B Concept 3 · transfer

The Moon is between Earth and the Sun. What phase is seen from Earth?

Target word: phase from position **Decides:** The lit half faces away from Earth.

A	Absolute language	Eclipses need exact alignment; most new moons pass above or below the Sun.
B	Correct	The sunlit side faces the Sun, away from Earth.
C	Adjacent category	First quarter is a right-angle position.
D	Inverted operation	Full moon occurs when Earth is between the Sun and Moon.

6. Answer D Concept 4

For a unit on plate tectonics, a teacher ~~with a popular rock collection~~ writes a big-idea statement: “Plates move because ...”. Which completion is accurate for a secondary course?

Target word: driving force of plates **Decides:** Internal heat and gravity.

A	Inverted operation	Earthquakes are a result of plate motion, not its cause.
B	Confusion set	The mantle is mostly solid rock that flows slowly.
C	Adjacent category	Rotation is not a main driver of plate motion.
D	Correct	Convection and slab pull are the accepted drivers.

7. Answer D Concept 5

While designing a water-cycle lesson ~~for a school near a large reservoir~~, a teacher wants students to name the main energy source that drives evaporation and keeps the cycle running. ~~The reservoir supplies 200,000 people.~~ Which answer should the lesson build toward?

Target word: energy source of the water cycle **Decides:** Evaporation needs energy.

A	Camouflage	Human systems move small amounts; they do not run the cycle.
B	Adjacent category	Geothermal heat is far too small to drive the global cycle.
C	Confusion set	Tides move ocean water but do not evaporate it.
D	Correct	Solar heating evaporates water and drives the winds that move it.

8. Answer D Concept 6

A coastal district’s ocean-science elective, ~~which takes one field trip per semester~~, is scheduling a tide-pool visit to see the lowest low tides. ~~The trip coordinator has a boat license.~~ Which moon phase should the teacher choose?

Target word: phase for the largest range **Decides:** Spring tides.

A	Dropped element	Drops the Sun’s contribution.
B	Confusion set	Light has no tidal effect; new moon works equally well.
C	Inverted operation	Quarter moons bring neap tides with the smallest range.
D	Correct	Aligned pulls add, giving the highest highs and lowest lows.

9. Answer B Concept 6 • transfer

During a first-quarter moon, what kind of tides occur?

Target word: tides at quarter moon **Decides:** Sun and Moon at right angles.

A	Adjacent category	Moon distance and phase are separate cycles.
B	Correct	Right-angle pulls partly cancel.
C	Confusion set	Lighting does not affect the Moon's gravity.
D	Inverted operation	Spring tides need alignment.

10. Answer B Concept 7

A teacher, ~~using a star chart app the whole department likes~~, has students plot stars on an H-R diagram. A student notices that most plotted stars fall along a diagonal band. What should the teacher help students conclude?

Target word: meaning of the band **Decides:** Most stars are on the main sequence.

A	Confusion set	Stars stay near one spot on the main sequence; the band is many stars of different masses.
B	Correct	The main sequence is the longest stage of a star's life.
C	Inverted operation	On the main sequence, brighter stars are hotter.
D	Adjacent category	The band sorts stars by mass, not age.

11. Answer C Concept 7 • transfer

Compared with a main-sequence star at the lower right of the H-R diagram, a main-sequence star at the upper left is:

Target word: properties by position **Decides:** Upper left = hot and luminous.

A	Inverted operation	That describes the lower-right star.
B	Adjacent category	Main-sequence position reflects mass, not age.
C	Correct	Mass sets position along the main sequence.
D	Confusion set	Mixes properties; hotter main-sequence stars are more massive and shorter-lived.

12. Answer A Concept 8

A team of teachers ~~at a school with block scheduling~~ is sequencing a ninth-grade Earth science course. Students have not yet studied density or heat transfer. The team wants to open the year with plate tectonics ~~because a volcano documentary just came out~~. What is the best sequencing decision?

Target word: sequencing choice **Decides:** Plate motion depends on density and convection.

A	Correct	Prerequisite ideas come before concepts that depend on them.
B	Dropped element	Drops the prerequisite that plate motion depends on.
C	Adjacent category	Textbook order is a resource, not a check on prerequisites.
D	Absolute language	Engagement matters, but no single opener is always right.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	10	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Backward design	Plan a unit by first naming the learning goals, then the evidence (assessments) that will show them, and only then the activities.	Starting from favorite activities or a textbook order.	A planner holding activities before any goals or assessments are set.
2	Three-dimensional learning	A strong science lesson weaves together a science practice, a disciplinary core idea, and a crosscutting concept in one task.	Calling a lesson strong because it covers the content or uses technology.	A lesson description that names what students do, learn, and connect.
3	Cause of Moon phases	Half the Moon is always lit by the Sun; phases show how much of that lit half faces Earth as the Moon orbits. Earth's shadow matters only in a lunar eclipse.	"Earth's shadow causes the phases."	A student explanation that uses a shadow.
4	What drives plate motion	Heat escaping Earth's interior drives slow mantle convection, and gravity pulls cold, dense slabs down at subduction zones (slab pull); the mantle is mostly solid rock that flows slowly.	Plates as rafts on a liquid mantle, or earthquakes pushing plates.	A "plates move because ..." statement.
5	Energy that drives the water cycle	Solar energy drives evaporation and atmospheric circulation; gravity returns water as precipitation and runoff.	Crediting Earth's interior heat or the Moon.	A question about what keeps the water cycle running.
6	Spring and neap tides	Tides come from differences in the Moon's (and Sun's) pull across Earth; when Sun, Earth, and Moon line up (new and full moon), tidal range is greatest (spring tides); at quarter moons it is smallest (neap tides).	Ignoring the Sun, or thinking moonlight or Moon brightness matters.	A moon phase paired with tide range.
7	The main sequence	Most stars fall on the main sequence because stars spend most of their lives fusing hydrogen in their cores; along it, hotter stars are brighter, more massive, and shorter-lived.	Reading the main sequence as one star's path over time.	A plotted H-R diagram with a diagonal band.
8	Sequencing by prerequisites	Order a course so students learn the ideas they need (such as density and heat transfer) before the concepts that depend on them (such as plate motion).	Ordering units by excitement, habit, or the textbook alone.	A unit that depends on ideas students have not yet learned.

Course scope (public description): UG Secondary Earth Science Curriculum prepares students to develop and evaluate Earth Science curricula for secondary education by examining the foundational principles of Earth's processes, materials, and their interactions within the broader context of the hydrosphere, atmosphere, and the universe. The course delves into the relationships between the Earth, Sun, and Moon, the dynamics of the Earth's internal and external systems, and the characteristics and life cycles of stars and planets. Learners will explore the structure and behavior of Earth's components, as well as celestial phenomena, equipping them with the knowledge necessary to effectively teach these concepts in secondary education settings. This course lays the groundwork for understanding and conveying the complexities of Earth and space science, preparing learners for impactful instruction in the classroom. Through practical examples and interactive activities, students will enhance their ability to design effective curricula that foster student engagement and understanding in Earth Science.